If $f:\mathbb{R}\to\mathbb{R}$ satisfies Intermidate Value Property and for any rational $r\in\mathbb{Q}$, $f^{-1}[\{r\}]$ is closed, then f is Continuous. (IVT property: If $f(a)<c<f(b)$, then $\exists x\in(a,b)$ s.t $c=f(x)$)

Proof. Suppose that f is not Continuous map. Then, there exists a sequence $\{x_n\}$ which converges to x_0 but $f(x_n)$ not converges to $f(x_0)$. That is, there exists $\varepsilon>0$ s.t

there exist infinitely many $n_i\in\mathbb{N}$ satisfying $|f(x_n)-f(x_0)|\geq\varepsilon$. $\left(\Leftrightarrow \begin{matrix} f(x_0)\leq f(x_n)-\varepsilon<f(x_n) \\ \text{or} \\ f(x_0)\geq f(x_n)+\varepsilon>f(x_n)\end{matrix}\right)$

By the density of rationals, there exist two rationals $r^-\in(f(x_0)-\varepsilon, f(x_0))$ and $r^+\in(f(x_0), f(x_0)+\varepsilon)$.

if $f(x_0)<f(x_{n_i})$, then $f(x_0)<r^+<f(x_{n_i})$ and

if $f(x_{n_i})<f(x_0)$, then $f(x_{n_i})<r^-<f(x_0)$.

By the IVP, for each n_i, there exists $t_i\in(x_0,x_{n_i})\cup(x_{n_i},x_0)$ s.t $f(t_i)=r^+$ or r^-.

Since $x_{n_i}\to x_0$ as $i\to\infty$, $t_i\to x_0$ as $i\to\infty$.

For each $i\in\mathbb{N}$, $t_i\in f^{-1}[\{r^+\}]\cup f^{-1}[\{r^-\}]$, closed set, therefore the limit point x_0 satisfies

$$x_0\in f^{-1}[\{r^+\}]\cup f^{-1}[\{r^-\}].$$

That is, $f(x_0)=r^+$ or r^-, it is contradiction.

Note: Continuous preserves the Compactness, so, a continuous map $f: X \to \mathbb{R}$ on compact X has minimum and maximum, since $f[X]$ is closed bounded.

Darboux's Theorem.
Let $f: [a,b] \to \mathbb{R}$ be a differentiable map.
Then, the derivative f' satisfies the Intermidate Value Theorem.
Proof. Suppose that $f'(a) < f'(b)$. Let α s.t $f'(a) < \alpha < f'(b)$ be given.
Define $g(x) = f(x) - \alpha x \quad (x \in [a,b])$.

Clearly g is differentiable, and $g'(a) = f'(a) - \alpha < 0$, $g'(b) = f'(b) - \alpha > 0$.
And, since g is continuous, it has a minimum in $[a,b]$, for some c.
By the assumption, $c \neq a$ and $c \neq b$. And, since g has a minimum at c,
$g'(c) = f'(c) - \alpha = 0$. So proved ☐

Corollary. If $f: I \to \mathbb{R}$ has a Primitive, then $f[I]$ is an interval.

Note: If f is defined on an interval and cannot satisfy IVT,
then it has no Primitive.